

Session 2: Explain

Interrogating five models, three independent ways

QUEST · ICOTS-12 Pre-Conference Workshop

11:00 – 12:30

Brisbane, Australia

Where We Are

Time	Session	QUEST
9:00 – 10:30	Session 1: Question & Understand	Q, U
11:00 – 12:30	Session 2: Explain	E
1:15 – 3:00	Session 3: Story & Transform	S, T

This session has one job: turn five fitted models into evidence you can defend.

Session 2 Roadmap

- | | | |
|-----|---------------------------|--------------------------------|
| 2.1 | Training five models | <i>icarm</i> |
| 2.2 | How stable is a result? | <i>caret, cross-validation</i> |
| 2.3 | Explaining with icarm | <i>global + local</i> |
| 2.4 | A model-agnostic view | <i>DALEX</i> |
| 2.5 | Beyond icarm's own models | <i>caret + DALEX</i> |
| 2.6 | A second opinion | <i>LIME</i> |
| 2.7 | Open exploration | <i>modelStudio</i> |

Carried Over From Session 1

`train_bin` and `test_bin` — 295 and 100 students, Mathematics course, split honestly in Session 1 with `icarm_split()`.

Every result in this session is computed on `test_bin` — data no model in this room has seen.

Can we trust any of these models?

Not “how accurate are they” — *trust*. That needs more than one number.

PROBLEM

Five model families exist for this problem. Which one should this workshop actually use?

2.1 · One Interface, Five Models

METHOD

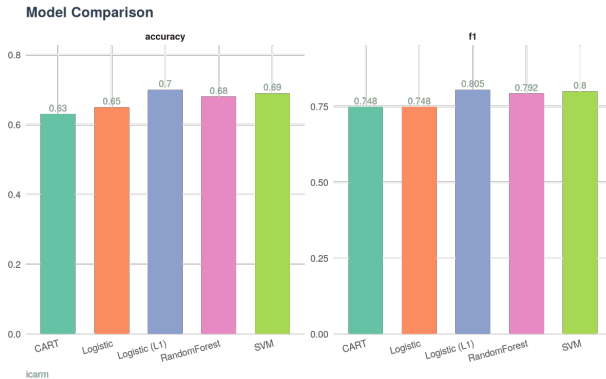
```
m_cart  <- icarm_fit(pass ~ ., train_bin, model = "cart",           positive =  
  "Pass")  
m_logit <- icarm_fit(pass ~ ., train_bin, model = "logistic",      positive =  
  "Pass")  
m_l1    <- icarm_fit(pass ~ ., train_bin, model = "logistic_l1",   positive =  
  "Pass")  
m_rf    <- icarm_fit(pass ~ ., train_bin, model = "random_forest", positive =  
  "Pass")  
m_svm   <- icarm_fit(pass ~ ., train_bin, model = "svm",           positive =  
  "Pass")
```

Three interpretable model families, two extended (“black-box”) — one function call each.

2.1 · Five Models, Compared

RESULT

VERIFIED



INSIGHT

The best-performing model here — sparse (L1) logistic regression, 0.70 accuracy — is fully interpretable. Random Forest (0.68) and SVM (0.69) are not meaningfully better, and are harder to explain.

Reaching for a black box is a decision, not a default. Here, the data doesn't reward it.

2.2 · How Stable Is a Result?

PROBLEM

A single train/test split gives exactly one number. How much would that number move under a different split?

2.2 · 5-Fold Cross-Validation

METHOD

```
ctrl <- caret::trainControl(method = "cv", number = 5,  
                           classProbs = TRUE, summaryFunction = twoClassSummary)  
cv_logit <- caret::train(pass ~ ., train_bin, method = "glm",  
                        family = "binomial", trControl = ctrl, metric = "ROC  
")
```

Refit the SAME model five times, each time holding out a different fifth of the training data.

2.2 · Five Folds, Five Numbers

RESULT

VERIFIED

Fold	ROC	Sens	Spec
1	0.593	0.316	0.700
2	0.670	0.300	0.900
3	0.674	0.263	0.875
4	0.755	0.450	0.897
5	0.744	0.632	0.769

INSIGHT

ROC ranges from **0.59** to **0.76** — entirely within the training data, before ever touching a different random seed for the split itself.

Every number in this session is a real number. None of them is THE number. Keep that in view as the session continues.

PROBLEM

We have a working model. We don't yet know what it's actually using to make its decisions — in general, or for any one student.

2.3 · Two Different Questions

METHOD

Global explanation

What does this model generally pay attention to, across everyone?

Local explanation

Why did it make THIS prediction, for THIS one student?

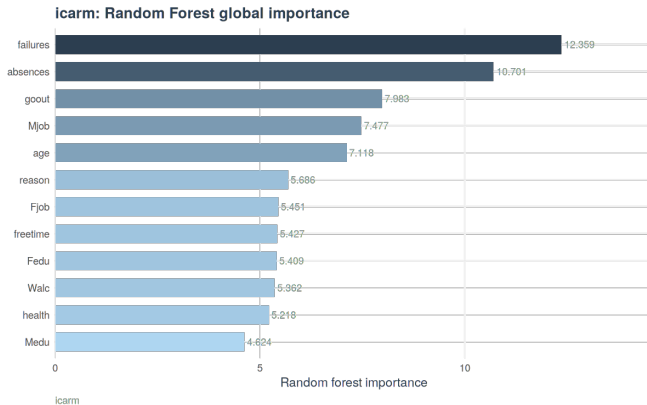
METHOD

```
ex_rf <- icarm_explain(m_rf, data = test_bin, label = "Random Forest")  
icarm_plot_importance(ex_rf, n_features = 12)
```


2.3 · What the Model Pays Attention To

RESULT

VERIFIED



METHOD

```
case_id <- which(predict(m_rf, test_bin, type = "class") == "Fail")[1]  
icarm_explain_local(ex_rf, newdata = test_bin[case_id, ], n_features = 6)
```

One row. One prediction. One explanation, specific to that student alone.

RESULT

VERIFIED

Three prior failures — the strongest single warning sign anywhere in the data.

Predicted Fail, 87% confidence.

INSIGHT

That student passed.

The model's single most confident rule, built from the strongest predictor in the whole dataset, was wrong for this real person. Global explanation tells you the pattern. Local explanation tells you its limits.

PROBLEM

`icarm_explain()` is built specifically for `icarm` models. What if you're handed a model `icarm` never fit?

METHOD

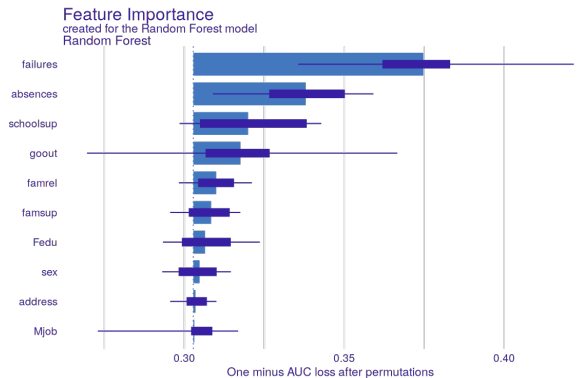
```
explainer_rf <- DALEX::explain(  
  m_rf$fit, data = test_bin, y = test_bin$pass == "Pass",  
  predict_function = function(m, d) predict(m, d, type = "prob")[, "Pass"],  
  label = "Random Forest", type = "classification")
```

One wrapper — model, data, and a predict_function — around anything.

2.4 · Permutation Importance

RESULT

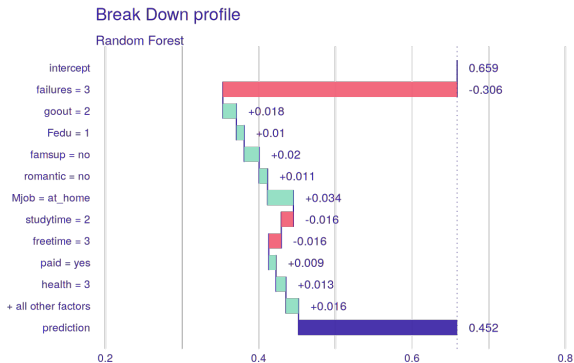
VERIFIED



2.4 · Break-Down for the Same Student

RESULT

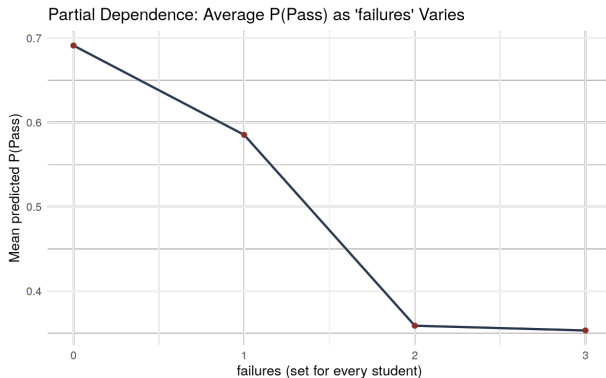
VERIFIED



2.4 · Partial Dependence

RESULT

VERIFIED



INSIGHT

DALEX's break-down and icarm's own local explanation agree on this student:
failures = 3 does almost all of the work.

Two different pieces of software, two different algorithms underneath, the same answer. That agreement is evidence — not proof, but evidence.

2.5 · Beyond icarm's Own Models

PROBLEM

Does DALEX only work on models icarm fit? Or does the “model-agnostic” claim actually hold?

2.5 · A Model icarm Never Touched

METHOD

```
gbm_caret <- caret::train(pass ~ ., train_bin, method = "gbm", verbose = FALSE
,
                        trControl = ctrl, metric = "ROC")
explainer_gbm <- DALEX::explain(
  gbm_caret, data = test_bin, y = test_bin$pass == "Pass",
  predict_function = function(m,d) predict(m,d,type="prob")[,"Pass"])
```

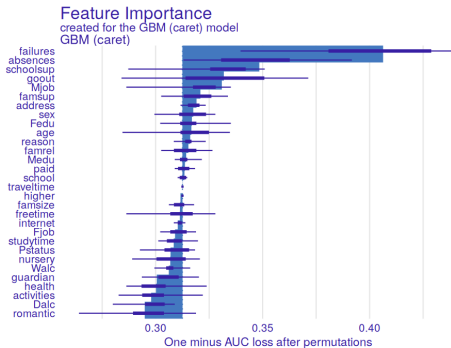
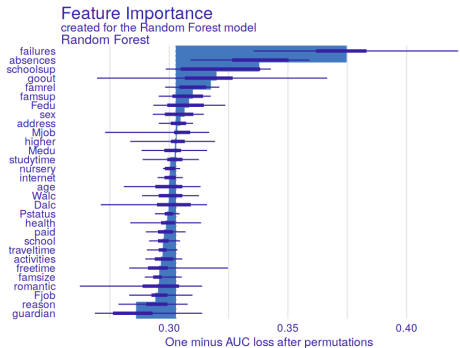
A gradient-boosted model, trained the standard way, handed to the exact same explanation machinery.

2.5 · Two Model Families, Side by Side

RESULT

VERIFIED

Global Importance: Random Forest vs. GBM (caret)



INSIGHT

GBM test accuracy: **0.69**. Random Forest: 0.68. Two entirely different algorithms — failures and absences on top in both.

icarm's five built-in models are a convenient starting set, not a ceiling. Everything from here forward works on any fitted model with a `predict()` method.

PROBLEM

DALEX's break-down is one explanation method. What if its particular mathematics is misleading us?

2.6 · A Completely Different Approach

METHOD

LIME explains one prediction by fitting a small, simple, interpretable model in the immediate neighbourhood of that single point — not by decomposing the original model at all.

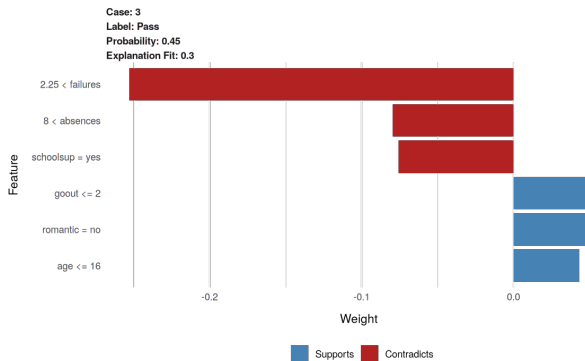
METHOD

```
lime_explainer <- lime::lime(train_bin, m_rf$fit)
lime_explanation <- lime::explain(test_bin[case_id, ], lime_explainer,
                                labels = "Pass", n_features = 6)
plot_features(lime_explanation)
```

2.6 · LIME's Explanation

RESULT

VERIFIED



INSIGHT

Compare this against DALEX's break-down two sections ago, for the same student.

Where two independent methods agree, trust grows. Where they disagree, that disagreement is information — not a bug to explain away.

METHOD

modelStudio combines break-down, importance, and partial dependence for MANY students into one interactive, offline dashboard.

2.7 · modelStudio, in Code

METHOD

```
ms <- modelStudio::modelStudio(  
  explainer_rf, new_observation = test_bin[1:6, ]  
ms  # opens in your browser
```

Take five minutes now. Pick a student. See if you can find something that surprises you.

Ninety Minutes, Three Explanation Methods

Method	How it works	On this data
icarm	Built into the fitting call	failures dominates
DALEX	Model-agnostic break-down	Agrees with icarm
LIME	A local model, refit each time	Compare for yourself

Accuracy is not enough.

Five models. Five-fold cross-validation. Three explanation methods. A model that got its single most confident prediction wrong. None of that shows up in a single accuracy number.

We now have models we can explain. Session 3 asks the harder questions:

Who does this model fail?

And what should
we do about it?

1:15 — Session 3

Story & Transform